

Task 1

1.1 Regulated Entity

The regulated entity selected for this study is The Bank of East Asia (BEA). Established in Hong Kong in 1918, BEA is one of the longest-standing and largest locally incorporated banks in Hong Kong, with its headquarters located in Central, Hong Kong. It operates branches and business outlets across multiple global regions. Its core businesses cover personal banking, corporate banking, wealth management, credit financing, and financial markets. With over a century of deep engagement in the Hong Kong market, BEA has also actively expanded its international operations. Among these, the U.S. market is a key region for its overseas expansion. BEA has branches and credit business units in cities such as New York and Los Angeles, specifically serving local Chinese communities and small and medium-sized enterprise (SME) clients by offering products such as residential mortgages, consumer loans, credit cards, and small business loans.

From a regulatory perspective, BEA operates under a dual-jurisdictional supervisory framework: In Hong Kong, as a licensed bank, BEA is fully regulated by the Hong Kong Monetary Authority (HKMA) and must strictly comply with a range of local regulatory rules, including the Banking Ordinance, the Code of Banking Practice for Personal Credit, the Guideline on Fair Trading and Consumer Protection, and the AI Risk Management Guideline issued in 2025. In the United States, its U.S. branches, as foreign bank branches, are subject to joint oversight by the Office of the Comptroller of the Currency (OCC), the Federal Reserve Board, and the Consumer Financial Protection Bureau (CFPB). They must strictly adhere to federal regulations such as the Equal Credit Opportunity Act (ECOA), Regulation B, the Fair Credit Reporting Act (FCRA), and the CFPB's 2025–2026 Guidance on AI-Powered Credit Denial Explanations, resulting in extremely stringent compliance requirements (Consumer Financial Protection Bureau, 2025,2026; Office of the Comptroller of the Currency, 2026).

1.2 Compliance Area

The compliance area selected for this study is algorithmic fairness and fair lending

compliance. The core reason for choosing this area is that, with the increasing digitalization of BEA's credit business, algorithmic models have been widely applied to key processes such as customer credit assessment, credit approval decision-making, risk pricing, and fraud detection. The fairness, transparency, and compliance of these algorithms directly impact customer rights, bank reputation, and cross-jurisdictional compliance risks. Moreover, this area exhibits the most significant regulatory differences and conflicts between the U.S. and Hong Kong, fully demonstrating the design value and application necessity of a "jurisdiction-aware" RegTech tool. It represents a critical pain point in compliance management for multinational financial institutions that urgently needs to be addressed (CFPB, 2025; Hong Kong Monetary Authority, 2025).

1.3 In-Depth Survey of Regulatory Differences between the U.S. and Hong Kong

(1) U.S. Regulatory Rules

The U.S. regulation of algorithmic fairness and fair lending is characterized by "strict anti-discrimination, mandatory explainability, rigid quantitative standards, and severe penalties for violations," forming one of the world's strictest credit fairness regulatory systems.

- **Core Legal Framework:** The Equal Credit Opportunity Act (ECOA) explicitly prohibits credit discrimination based on protected characteristics such as sex, race, age, religion, marital status, and national origin, covering both overt discrimination and indirect discriminatory effects (disparate impact). Regulation B, as the implementing rule of ECOA, further details compliance requirements for credit application, information disclosure, and adverse action notices. New rules issued by the CFPB in 2025–2026 reinforce the explainability obligations for AI algorithm-based credit denials, requiring institutions to provide applicants with clear, specific, understandable, and non-technical reasons for denial, clearly indicating the key factors influencing the algorithm's decision, and prohibiting the use of algorithmic automated decision-making as a pretext for refusing to provide explanations (CFPB, 2025, 2026).
- **Quantitative Compliance Standards:** U.S. regulators enforce strict quantitative thresholds to monitor algorithmic fairness. For example, the difference in credit approval rates

between protected groups (e.g., women, minorities) and non-protected groups must not exceed 10%. Any exceedance is deemed to constitute disparate impact, triggering immediate algorithmic audits, model adjustments, and compliance remediation; otherwise, penalties may follow (CFPB, 2026).

- **Consequences of Violations:** Financial institutions that violate fair lending rules face severe consequences, including high civil penalties (up to millions or tens of millions of dollars), mandatory regulatory remediation, business restrictions, class-action lawsuit settlements, and serious reputational damage. Historically, several major banks have been heavily fined by the CFPB for algorithmic discrimination issues (CFPB, 2025).

(2) Hong Kong Regulatory Rules

Hong Kong's regulation of algorithmic fairness and fair lending is characterized by "consumer protection priority, outcome-oriented approach, human oversight as a backstop, and moderately flexible rules." While equally emphasizing fair lending, Hong Kong has not adopted the rigid quantitative standards of the U.S., and its supervisory logic is more tailored to the realities of the local financial market.

- **Core Regulatory Guidance:** The HKMA's Code of Banking Practice for Personal Credit requires banks to follow the principles of fairness, impartiality, and transparency in credit operations, to refrain from unreasonable discrimination against customers, and to protect the legitimate rights and interests of vulnerable customers (e.g., the elderly, low-income groups). The AI Risk Management Guideline issued in 2025 imposes, for the first time, specific regulatory requirements for financial AI applications, emphasizing that banks must establish AI algorithm governance frameworks to ensure algorithmic decisions are explainable, auditable, fair, and reasonable, but without mandating uniform quantitative fairness metrics (Hong Kong Monetary Authority, 2025).
- **Core Compliance Requirements:** Hong Kong regulation focuses more on outcome fairness and process transparency. It requires banks to ensure that credit approval outcomes do not systematically and unfairly exclude specific groups, to maintain complete approval records and algorithmic decision logs subject to supervisory spot checks, and to emphasize the irreplaceability of human oversight. Algorithms may only

serve as decision-support tools; final credit approval decisions must be made by qualified human reviewers, and fully automated algorithmic decision-making is prohibited (Hong Kong Monetary Authority, 2025).

- **Quantitative Standards and Explanation Requirements:** Hong Kong regulation does not prescribe rigid approval rate difference thresholds like those in the U.S. Algorithmic fairness assessments are primarily conducted through case-by-case reviews and outcome-based judgments. Generally, an approval rate difference between groups of up to 18% is considered within a reasonable range. There is no mandatory obligation to provide explanations for credit denials; banks are only required to provide simple and understandable reasons for approval decisions upon customer inquiry, without needing to issue detailed technical explanation reports (Hong Kong Monetary Authority, n.d.).
- **Consequences of Violations:** Penalties for regulatory violations in Hong Kong are relatively mild. They primarily include regulatory warnings, remediation orders, internal accountability, and public reprimands. Fine amounts are considerably lower than those in the U.S. The core objective is to prompt banks to rectify improper conduct and protect consumer rights, rather than to impose pure punishment (Hong Kong Monetary Authority, n.d.).

1.4 Summary of U.S.-Hong Kong Regulatory Differences

In summary, the regulatory differences between the U.S. and Hong Kong in the area of algorithmic fairness and fair lending compliance fundamentally represent a divergence between "rigid rules vs. flexible governance, process compliance vs. outcome compliance, quantitative mandates vs. qualitative assessment, high penalties vs. low intervention." This divergence directly results in BEA facing two entirely different sets of compliance standards for its cross-jurisdictional credit operations. Traditional manual approaches cannot simultaneously satisfy the requirements of both jurisdictions, creating an urgent need for technical tools to achieve rule adaptation and automated compliance management (CFPB, 2025, 2026; Hong Kong Monetary Authority, 2025; OCC, 2026).

References

Consumer Financial Protection Bureau. (2025). Guidance on AI-powered credit denials explanations.

<https://www.consumerfinance.gov/about-us/newsroom/cfpb-issues-guidance-on-credit-denials-by-lenders-using-artificial-intelligence/>

Consumer Financial Protection Bureau. (2026). Regulation B (12 CFR Part 1002).

https://files.consumerfinance.gov/f/documents/cfpb_reg-b_fr-notice-proposed-rule_2025-11.pdf
files.consumerfinance.gov

Hong Kong Monetary Authority. (2025). AI risk management guideline.

<https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20241107-1-EN/20241107-1-EN.pdf>
[Hong Kong Monetary Authority](https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20241107-1-EN/20241107-1-EN.pdf)

Hong Kong Monetary Authority. (n.d.). Code of banking practice for personal credit.

<https://www.hkma.gov.hk/eng/key-information/guidelines-and-circulars/>

Office of the Comptroller of the Currency. (2026). OCC Bulletin 2026-13: Artificial intelligence risk management.

[https://www.occ.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html](https://www OCC.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html)

Board of Governors of the Federal Reserve System. (n.d.). Fair Credit Reporting Act (FCRA).

https://www.federalreserve.gov/consumers_credit.htm